Chemistry Important Facts

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Introduction

1. Acids - Substances that produce hydrogen ions (H^+) in water and have a sour taste.

- **Examples:** Hydrochloric acid (HCl), Sulphuric acid (H_2SO_4), Citric acid (in lemons).

2. Bases- Substances that produce hydroxide ions (OH^-) in water and have a bitter taste or soapy feel.

- **Examples:** Sodium hydroxide (NaOH), Calcium hydroxide (lime water), Ammonium hydroxide (NH_4OH).

3. Salts - Compounds formed by the neutralization of an acid with a base.

- **Examples:** Sodium chloride (table salt – NaCl), Potassium nitrate (KNO_3), Calcium carbonate ($CaCO_3$).

4. Petroleum Products- Products derived from crude oil through refining processes.

- **Examples:** Petrol, diesel, kerosene, LPG (Liquefied Petroleum Gas).

5. Fertilizers - Chemical or natural substances that provide nutrients to plants and improve soil fertility.

- **Examples:** Urea (Nitrogen fertilizer), Superphosphate (Phosphorus fertilizer), Potash (Potassium fertilizer).

6. Pesticides- Chemicals used to kill or control pests, insects, or weeds affecting crops.

- **Examples:** DDT (insecticide), Malathion (insecticide), 2,4-D (weed killer).

7. Metallurgy - The science of extracting metals from ores, refining them, and preparing alloys.

- **Examples:** Extraction of iron from hematite, copper from chalcopyrite, and aluminum from bauxite.

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Director,

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Acids present in fruits and vegetables

Fruits Acid Present

Apples	Malic acid
Apricots	Malic acid and Citric acid
Avocados	Tartaric acid
Banana	Malic acid
Blackberries	Iso citric acid
Cranberries	Citric acid
Figs	Citric acid
Gooseberries	Citric acid
Grapes	Malic acid and Tartaric acid
Kumquat	Citric acid
Lemons	Citric acid
Orange	Citric acid
Passion Fruit	Malic acid
Pears	Malic acid
Pineapples	Citric acid
Raspberry	Citric acid
Rosehip	Malic acid
Strawberries	Citric acid

Vegetables Acid Present

Beans	Citric acid
Broccoli	Malic acid
Mushrooms	Lactarimic acid
Potatoes	Malic acid
Rhubarb	Malic acid
Tomatoes	Malic acid

Properties of Acid Base indicator

Theories of acid base indicator

Arrhenius theory

The **Arrhenius theory** of acids and bases states that "an acid generates H^+ ions in a solution whereas a base produces an OH^- ion in its solution".

Bronsted-Lowry theory

The **Bronsted-Lowry theory** defines "an acid as a proton donor and a base as a proton acceptor"

Lewis theory

The **Lewis definition** of acids and bases describes "acids as electron-pair acceptors and bases as electron-pair donors"

Properties

Properties of Acids

- Acids are corrosive in nature.
- They are good conductors of electricity.
- Their pH values are always less than 7.
- When reacted with metals, these substances produce hydrogen gas.
- Acids are sour in taste.
- Examples: Sulfuric acid [H_2SO_4],Acetic acid [CH_3COOH]

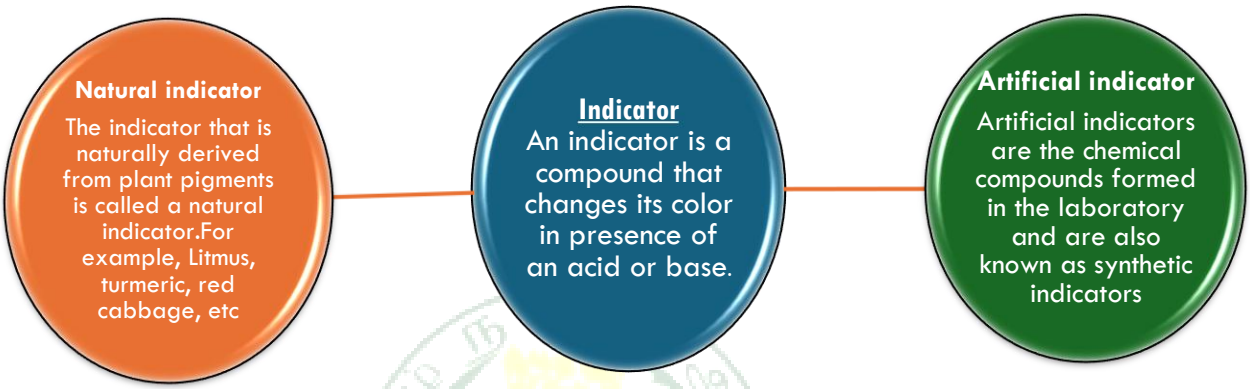
Properties of Bases

- Some properties, like a bitter taste, are owned by all bases.
- They are found to have a soapy texture when touched.
- These substances release hydroxide ions (OH^- ions) when dissolved in water.
- In their aqueous solutions, bases act as good conductors of electricity.
- The pH values corresponding to bases are always greater than 7.
- Bases are bitter-tasting substances which have the ability to turn red litmus paper blue.
- Examples: Sodium hydroxide [NaOH], milk of magnesia [$Mg(OH)_2$], calcium hydroxide [$Ca(OH)_2$].

Neutral Substances

- These substances do not display any acidic or basic characteristics.
- Their pH values approximate to 7.
- Neutral substances have no effect on red or blue litmus paper.
- The pH of pure water is exactly 7.
- Examples: Water, Common salt (NaCl)

Indicator



Name	Acid Color	pH Range of Color Change	Base Color
Methyl violet	Yellow	0.0 - 1.6	Blue
Thymol blue	Red	1.2 - 2.8	Yellow
Methyl orange	Red	3.2 - 4.4	Yellow
Bromocresol green	Yellow	3.8 - 5.4	Blue
Methyl red	Red	4.8 - 6.0	Yellow
Litmus	Red	5.0 - 8.0	Blue
Bromothymol blue	Yellow	6.0 - 7.6	Blue
Thymol blue	Yellow	8.0 - 9.6	Blue
Phenolphthalein	Colorless	8.2 - 10.0	Pink
Thymolphthalein	Colorless	9.4 - 10.6	Blue

Alizarin yellow R	Yellow	10.1 - 12.0	Red

Important pH and their values

pH VALUE:

- Historically pH denotes the "potential of hydrogen"
- pH is a measure of how acidic/basic water is
- The range goes from 0 - 14, with 7 being neutral. pHs of less than 7 indicate acidity, whereas a pH of greater than 7 indicates a base

pH Values List:

Substances	pH Value
Battery acid	1.0
Gastric juice	1.0 – 3.5
Soft drink	2.0 – 4.0
Lemon Juice	2.2 – 2.4
Vinegar	2.4
Wine	2.8
Lime	2.8
Fruit jellies	2.8- 3.4
Apple juice	2.9 – 3.3
Strawberries	3.0- 3.5
Soda Water	3-4
Orange juice	3.7
Tomato juice	4.0
Butter	5
Coffee	5
Yoghurt	4- 4.6
Curd	4.5-5.5
Beer	4.5
Urine	4.5-8.0

Cheese	5.1-5.9
Tea	5.5
Acid Rain	< 5.5
Peas	5.8 – 6.4
Urine(human)	6.0
Soft drink	6.0
Fresh Milk	6.5- 6.7
Cow Milk	6.4-6.8
Buffalo Milk	6.7-6.8
Saliva	6.2-7.6
Pure Drinking Water	7
Salt	7
Copper Sulphate	Greater than 7
Sugar	More or less than 7
Blood plasma	7.35-7.45
Alcohol	7.33
Blood	7.4
Eggs	7.8
Toothpaste	8.0
Baking Soda Solution	8.5
Washing Soda Solution	9.0
Oxygen	7.4
Soap	7.5-9
Baking soda	9
Oil	9.0-11.0
Milk of magnesia	10.5
Ammonia	11
Bleaching Powder	11
Sodium Hydroxide	13
Concrete	13

Periodic classification facts

- Elements existing in nature - **88**
- Artificially produced elements - **24**
- Number of metallic elements - **90**
- Number of non metallic elements - **22**
- The element to be found most abundantly on the earth's surface - **Oxygen**
- The metallic element to be found most abundantly on the earth's surface - **Aluminium**
- The lightest element to be found in nature - **Hydrogen**
- The heaviest element to be found in the nature - **Osmium**
- The lightest metallic element to be found in nature - **Lithium**
- The lightest non-metallic element to be found in nature - **Hydrogen**
- The liquid metal - **Mercury**
- The liquid non metal - **Bromine**
- The metal which is the best conductor of the electricity - **Silver**
- The most malleable element to be found in nature - **Gold**
- The maximum electrons acceptable by an element in nature - **Chlorine**
- The group of most gaseous elements - **Zero group**
- **Hydrogen (**H) is the first element and **Oganesson** (O) is the last element (Og).
- The first Periodic Table was produced by Russian chemist **Dmitri Mendeleev.**
- The Periodic Table is controlled by the International Union of Pure Applied Chemistry **IUPAC.**
- The element that is most prevalent in the universe is **hydrogen (H).** It accounts for roughly 75% of the entire observable cosmos.

USES OF RADIO ISOTOPES

Radioactive isotope

Radioactive isotope, any of several species of the same chemical element with different masses whose nuclei are unstable and dissipate excess energy by spontaneously emitting radiation in the form of alpha, beta, and gamma rays

Isotope and their Uses:

Americum-241	Many smoke detectors for homes and businesses use this material to assess toxic lead levels in dried paint samples to ensure uniform thickness in rolling processes such as steel and paper manufacturing as well as to assist in determining where oil wells should be drilled.
Cadmium-109	Analyzes metal alloys for stock control and scrap sorting.
Calcium-47	Biomedical researchers studying cellular functions and bone formation in mammals will benefit greatly from this discovery.
Californium-252	Used to inspect airline luggage for hidden explosives, to determine the moisture content of soil in the road and building industries, and to measure the moisture of materials stored in soils.
Carbon-14	Important research tool. Aids in research by ensuring that potential new drugs are metabolised without the formation

	of harmful by-products. Biological research, agriculture, pollution control, and archaeology all use it.
Caesium-137	Used to treat cancerous tumours, to measure correct patient doses of radioactive pharmaceuticals, to measure and control liquid flow in oil pipelines, to tell researchers whether oil wells are plugged by sand, and to ensure the proper fill level for packages of food, drugs, and other products.
Chromium-51	Used in studies of red blood cell survival.
Cobalt-57	As a tracer, it is used to diagnose pernicious anaemia.
Cobalt-60	Sterilizes surgical instruments as well as improves the safety and reliability of industrial fuel oil burners. Cancer treatment, food irradiation, gauges, and radiography all use it.
Copper-67	When monoclonal antibodies are injected into a cancer patient, it aids the antibodies in binding to and destroying the tumour.
Curium-244	In mining, it is used to analyse material excavated from pits as well as slurries from drilling operations.
Gallium-67	Used to make medical diagnoses
Iodine-123	It is commonly used to diagnose Thyroid disorders and other metabolic disorders, including brain function.
Iodine-125	A significant diagnostic tool used in clinical tests and to diagnose thyroid disorders. This term is also used in biomedical research.
Iodine-131	Thyroid disorders are treated with this medication. (Disease of Graves)
Iridium-192	It is used to test the integrity of pipeline welds, boilers, and aircraft parts, as well as in brachy therapy/tumour irradiation.
Iron-55	Detects the presence of sulphur in the air and analyses electroplating solutions. It is used in metabolic research.
Krypton-85	• Indicator lights in appliances such as clothes washers and dryers, stereos, and coffeemakers are used to determine the thickness of thin plastics and sheet metal, rubber, textiles, and paper, as well as to measure dust and pollutant levels.
	• Used in explosive detection, voltage regulators and current surge protectors in electronic devices, and electron capture detectors in gas chromatographs.

Nickel-63	Used in explosive detection, voltage regulators and current surge protectors in electronic devices, and electron capture detectors in gas chromatographs.
Phosphorus-32	This term is used in molecular biology and genetics research.
Phosphorus-33	Since 1972, it has powered more than 20 NASA spacecraft.
Plutonium-238	Reduces static charge in the manufacture of photographic film and other materials.
Polonium-210	Used to measure the thickness of thin plastics, thin sheet metal, rubber, textiles, and paper in electric blanket thermostats.
Promethium-147	Used to measure the thickness of thin plastics, thin sheet metal, rubber, textiles, and paper in electric blanket thermostats.
Radium-226	Increases the effectiveness of lighting rods
Selenium-75	In life science research, it is used in protein studies.
Sodium-24	Leak detection in industrial pipe lines and oil well studies.
Strontium-85	It is used to research bone formation and metabolism.
Strontium-90	Survey metres are used in schools, the military, and emergency management authorities. Sensors and medical treatment are also applications for this material.
Sulphur-35	This term is used in genetics and molecular biology research.
Technetium-99m	The most commonly used radioactive pharmaceutical in nuclear medicine diagnostic studies. For brain, bone, liver, spleen, and kidney imaging, as well as blood flow studies, various chemical forms are used.
Thallium-201	In nuclear medicine, it is used for nuclear cardiology and tumour detection.
Thallium-204	Dust and pollutant levels on filter paper are measured, and the thickness of plastics, sheet metal, rubber, textiles, and paper is gauged.
Thoriated Tungsten	Electric arc welding rods are used in the construction, aircraft, petrochemical, and food processing industries. They result in easier starting, improved arc stability, and reduced metal contamination.
Thorium-229	It extends the life of fluorescent lights.
Thorium-230	Colour and fluorescence are provided in coloured glazes and glassware.

Tritium A	vital tool in biomedical research. Used in life science and drug metabolism research to ensure the safety of potential new drugs for self-illuminating aircraft and commercial exit signs for luminous dials, gauges, and wrist watches, as well as geological prospecting and hydrology.
Uranium-234	Used to provide a natural colour and brightness in dental fixtures such as crowns and dentures.
Uranium-235	It is used to make fluorescent glassware, a variety of coloured glazes, and wall tiles, as well as fuel for nuclear power plants and naval nuclear propulsion systems.
Xenon-133	In nuclear medicine, it is used to study lung ventilation and blood flow.

COLLOIDAL SOLUTION

COLLOIDAL SOLUTION:

Dispersion Medium	Dispersed Phase	Type of Colloid	Example
Solid	Solid	Solid sol	Ruby glass
Solid	Liquid	Solid emulsion/gel	Pearl, cheese
Solid	Gas	Solid foam	Lava, pumice
Liquid	Solid	Sol	Paints, cell fluids
Liquid	Liquid	Emulsion	Milk, oil in water
Liquid	Gas	Foam	Soap suds, whipped cream
Gas	Solid	Aerosol	Smoke

Gas	Liquid	Aerosol	Fog, mist

Metal,Non metal and commercial uses

Metal:

- These are the elements which are hard, lustrous, ductile, malleable, sonorous and conductor of heat and electricity in their solid as well as molten state.

Non metal:

- These may be solid, liquid or gas (bromine is the only liquid non-metal).

Elements	Uses
Carbon dioxide (CO2)	Soda water, Fire extinguisher
In phosgene gas (COCl2)	Carbon monoxide (CO)
Phosphorous	Red (P4) as refrigerant, match industry, white (P4) as rat killing medicine
Graphite	As electrodes
Diamond	Ornaments, Glass cutting, rock drilling
Alum [K2SO4 Al2 (SO4)3. 24 H2O]	Purification of water and leather industry
Aluminium sulphate (Al2(SO4)3. 18H2O)	In paper industry, fire extinguisher
Anhydrous Aluminium chloride (AlCl3)	Cracking of petroleum

Mercuric chloride (HgCl2)	Calomel, insecticides (corrosive sublimate)
Mercuric oxide (HgO)	Ointment, Poison
Mercury (Hg)	Thermometer vermillion, amalgam
Zinc Sulphide (ZnS)	White pigment
Zinc sulphate (White vitriol) (ZnSO4.7H2O)	Lithopone, eye ointment
Zinc chloride (ZnCl2)	Textile industry
Zinc oxide (ZnO)	Ointment
Zinc (Zn)	In battery
Calcium carbide (CaC2)	Calcium cyanide and acetylene gas
Bleaching Powder (Ca(OCl)2)	Insecticides, bleaching actionsPlaster of Paris Statue, surgery
Calcium sulphate (CaSO4.2H2O)	Cement industry
Calcium carbonate (CaCO3)	Lime and toothpaste
Carbon dioxide (CO2)	Soda water, fire extinguisher
Carbon monoxide (CO)	In phosgene gas (COCl2)
Graphite	As electrodes
Copper sulphate (CuSO4.5H2O)	Insecticides, electric cells
Cupric oxide (CuO)	Blue and green glass, purification of petroleum
Cuprous Oxide (Cu2O)	Red glass, pesticides
Copper (Cu)	Electrical wire
Sodium nitrate (NaNO3)	Fertilizer
Sodium sulphate (Glauber salt)	Medicine, cheap glass
Sodium bicarbonate (Baking soda)	Fire extinguisher, bakery for preparing soft bread,reagent

Sodium carbonate (Washing soda)	Glass industry, paper industries, removal of permanent hardness of water
Hydrogen peroxide	Oxidants and reductants, insecticides
Liquid hydrogen	Rocket fuel

Important Alloy

Alloys And their uses:

Alloys	Compositions	Uses
Brass	Cu (70%) + Zn (30%)	**In making utensils**
Bronze	Cu (90%) + Sn (10%)	**In making coins, bell and utensils**
German Silver	Cu + Zn + Ni (60% + 20% + 20%)	**In making utensils**
Rolled Gold	Cu (90%) + Al (10%)	**In making cheap ornaments**
Gun Metal	Cu + Sn + Zn + Pb (88% + 10% + 1% + 1%)	**In making gun, barrels, gears and bearings**
Delta Metal	Cu + Zn + Fe (60% + 38% + 2%)	**In making blades of aero plane**

Muntz Metal	Cu (60%) + Zn (40%)	**In making coins**
Dutch Metal	Cu (80%) + Zn (20%)	**In making artificial ornaments**
Monel Metal	Cu (70%) + Ni (30%)	**For base containing container**
Rose Metal	Bi + Pb + Sn (50% + 28% + 22%)	**For making automatic fuse**
Solder	Pb (50%) + Sn (50%)	**For soldering**
Magnalium	Al (95%) + Mg (5%)	**For the frame of aeroplane**
Duralumin	Al + Cu + Mg + Mn (94% + 3% + 2% + 1%)	**For making utensils**
Type Metal	Sn + Pb + Sb (5% + 80% + 15%)	**In printing industry**
Bell Metal	Cu (80%) + Sn (20%)	**For casting bells and statues**
Stainless steel	Fe + Cr + Ni + C (75% + 15% + 10% + 0.05%)	**For making utensils and surgical cutlery**

Nickel steel	**Fe (95%) + Ni (5%)**	**For making electrical wire and automobile parts**

Important ores

Name of Elements	Ores	Chemical Formula
1. Ores of Aluminium (Al)	(i) Bauxite (ii) Corundum (iii) Cryolite (iv) Feldspar (v) Alunite (vi) Kaolin or Kaolinite	$Al_2O_3.2H_2O$ Al_2O_3 Na_3AlF_6 $KAlSi_3O_8$ $KAl_3(SO_4)_2(OH)_6$ $Al_2O_3.2SiO_2.2H_2O$
2. Ores of Iron (Fe)	(i) Hematite (ii) Magnetite (iii) Iron Pyrite (iv) Siderite (v) Limonite	Fe_2O_3 Fe_3O_4 FeS_2 $FeCO_3$ $Fe_2O_3.3H_2O$
3. Ores of Copper (Cu)	(i) Copper Pyrite or Chalco pyrite (ii) Copper Glance (iii) Cuprite (iv) Malachite	$CuFeS_2$ Cu_2S Cu_2O $CuCO_3.Cu(OH)_2$
4. Ores of Zinc (Zn)	(i) Zinc Blend or Sphalerite (ii) Smithsonite or Calamine (iii) Zincite	ZnS $ZnCO_3$ ZnO
5. Ores of Sodium (Na)	(i) Rock Salt (ii) Sodium Carbonate	NaCl Na_2CO_3
6. Ores of Potassium (K)	(i) Carnalite (ii) Saltpetre	$KCl.MgCl.6H_2O$ KNO_3

7. Ores of Lead (Pb)	(i) Galena (ii) Anglesite	PbS $PbSO_4$
8. Ores of Tin (Sn)	(i) Tin Pyrites (ii) Cassiterite	Cu_2FeSnS_4 SnO_2
9. Ores of Silver (Ag)	(i) Silver Glance or Argentite (ii) Chlorargyrite	Ag_2S AgCl
10. Ores of Gold (Au)	(i) Calaverite or Gold Telluride (ii) Sylvanite	$AuTe_2$ $(Ag,Au)Te_2$
11. Ores of Mercury (Hg)	(i) Cinnabar (ii) Calomel	HgS Hg_2Cl_2
12. Ores of Magnesium (Mg)	(i) Dolomite (ii) Carnallite (iii) Magnesite	$CaMg(CO_3)_2$ $KMgCl_3.6H_2O$ $MgCO_3$
13. Ores of Calcium (Ca)	(i) Limestone (ii) Dolomite (iii) Gypsum (iv) Anhydrite	$CaCO_3$ $CaMg(CO_3)_2$ $CaSO_4.2H_2O$ $CaSO_4$
14. Ores of Phosphorous (P)	(i) Phosphorite (ii) Fluorapatite	$Ca_5(PO_4)_3F$ $Ca_5(PO_4)_3F$
15. Ores of Manganese (Mn)	(i) Pyrolusite (ii) Mangnite (iii) Braunite	MnO_2 $Mn_2O_3.H_2O$ $3Mn_2O_3$, $MnSiO_3$

Catalyst and its uses

CATALYSTS AND ITS USES

Catalysts	Uses
Fe + Mo	Synthesis of NH3 by Haber's process
Ni	Synthesis of vanaspati ghee (hydrogenation)
Pt	Synthesis of H2SO4 by the contact process

NO	Manufacture of H2SO4 by the lead chamber process
Hot Al2O	Preparation of Ether from alcohol
CuCl2	Preparation of chlorine gas by Deacon process

CATALYSTS AND ITS PROCEES:

S.No.	*Process*	*Catalysts*
1.	Manufacture of ghee from vegetable oils	Nickel
2.	Contact process for the manufacture of sulphuric acid	Pt powder
3.	Decon's process for the manufacture of chlorine	Cupric chloride
4.	Conversion of proteins into peptide	Pepsin enzyme
5.	Conversion of glucose into ethyl alcohol	Zymase enzyme
6.	Conversion of starch into maltose	Diastase enzyme
7.	Formation of vinegar from sugarcane	Micoderma aceti
8.	Conversion of sucrose into glucose and fructose	Invertase enzyme
9.	Conversion of milk into curd	Lactase (Lactobacilli)

Fuel gas

Fuel	*Composition*		*Sources*
Water gas	Carbon monoxide (CO) + Hydrogen(H_2)		By passing steam over red hot coke
Producer gas	Carbon monoxide(CO)+ Nitrogen(N_2)		Bypassing insufficient air over red hot coke
Coal gas	Hydrogen(H_2)+Methane(CH_4) +Ethylene+Acetylene+CO		By fractional distillation of wood
Biogas or Gobar gas	Methane (CH_4) + Carbon dioxide (CO_2)+ Hydrogen (H2) +Nitrogen (N2)		Fromorganicwastes
Hydrogen Fuel	Dihydrogen(H_2)		In fuel cell for generating electric power

Various grades of coal

TYPES OF COAL:

- There are four major types (or "ranks") of coal

Anthracite
The highest rank of coal. It is a hard, brittle, and black lustrous coal, often referred to as hard coal, containing a high percentage of fixed carbon and a low percentage of volatile matter

Bituminous
Bituminous coal is a middle rank coal between subbituminous and anthracite. Bituminous coal usually has a high heating (Btu) value and is used in electricity generation and steel making in the United States. Bituminous coal is blocky and appears shiny and smooth when you first see it, but look closer and you might see it has thin, alternating, shiny and dull layers.

Subbituminous
Subbituminous coal is black in color and is mainly dull (not shiny). Subbituminous coal has low-to-moderate heating values and is mainly used in electricity generation.

Lignite
Lignite coal, aka brown coal, is the lowest grade coal with the least concentration of carbon. Lignite has a low heating value and a high moisture content and is mainly used in electricity generation

Coal types And Properties

Coal types	Properties
Bituminous	Black, soft, smoky, flame, domestic fuel
Lignite	High moisture content burns easily, low calorific value
Peat	Low-grade coal produces less heat, more smoke and ash

Anthracite	Superior quality, hardest form, high calorific value, high carbon content

Common names and their compound

Common Name	Chemical Compounds/Elements	Chemical Formula	Uses
Baking Soda	Sodium Hydrogen Carbonate or Sodium bicarbonate	$NaHCO_3$	It is used as an ingredient in baking.
Antiformin/ Bleach/ Chloride of Soda	Sodium hypochlorite	NaOCl/NaClO	It is used on a large scale for surface purification, bleaching, odour removal and water disinfection.
Bleaching Powder	Calcium Oxychloride	$CaOCl_2$	It is used as a bleaching agent and disinfectant.
Blue Vitriol	Copper Sulphate	$CuSO_4.xH_2O$	It is used to control fungus and algae.
Borax	Sodium tetraborate decahydrate/ Sodium Borate	$Na_2[B_4O_5(OH)_4].8H_2O$	It is used in laundry detergent.
Calomel	Mercurous chloride	Hg_2Cl_2	It is used as a fungicide, and formerly in medicine as a purgative.
Carbolic Acid/ Hydroxybenzene/ Phenic Acid	Phenol	C_6H_5OH	It is used as a disinfectant in household cleaners.
Carborundum	Silicon carbide	SiC	It is used in automotive brakes and clutches.
Caustic potash	Potassium hydroxide	KOH	It is used in liquid fertilizers, potassium soaps and detergents.
Chile Saltpetre	Sodium Nitrate	$NaNO_3$	It is used for making high-strength glasses, fertilizers and explosives.

Chloroform	Trichloromethane	$CHCl_3$	It is used as a solvent for lacquer, floor polishes and resins.
Cream of tartar	Potassium hydrogen tartrate/ potassium bitartrate	$KC_4H_5O_6$	It is used in bakery to stabilize egg whites.
Dolomite	Calcium magnesium carbonate	$CaMg(CO_3)_2$	It is used as an ingredient in the production of glass, bricks, and ceramics.
Dry Ice	Solid carbon dioxide	CO_2	It is used to preserve food.
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4.7H_2O$	It is known to relieve pain and inflammation.
Freon-12/CFC-12	Dichlorodifluoromethane	CCl_2F_2	It is used as a refrigerant in refrigerators and air conditioners.
Galena	Lead (II) sulphide	PbS	It is also used in a green glaze applied to pottery.
Glauber's Salt	Sodium sulphate decahydrate	$Na_2 SO_4.10H_2O$	It is used in medicine and as a dyeing agent.
Glycerol	Propane-1,2,3-triol	$C_3H_8O_3$	It is used in food and skin care industry.
Green Vitriol	Ferrous sulphate heptahydrate	$FeSO_4. 7H_2O$	It is used in sewage and water treatment, and as a pigment, fertilizer.
Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$	It is used to manufacture dry wall.
Heavy water	Deuterium Oxide	D2O	It is used in nuclear reactors, as a neutron moderator.
Hypo Solution	Sodium Thiosulphate pentahydrate	Na2S2O3.5H2O	It is used for both film and photographic paper processing.
Hematite	Iron (III) Oxide/ Ferric Oxide	Fe2O3	It is used to produce pigments, preparations for heavy media separation, radiation shielding etc.

Laughing gas	Dinitrogen monoxide / Nitrous Oxide	N_2O	It is used by dentist.
Quick Lime/ Burnt Lime	Calcium Oxide	CaO	It is used in production of iron and steel, paper and pulp and in treatment of water.
Slaked Lime	Calcium Hydroxide	$Ca(OH)_2$	It is used in mortars, plasters, and cements.
Limestone	Calcium carbonate	$CaCO_3$	It is used as a building material.
Litharge(Red) and Massicot (Yellow)	Lead (II) Oxide/ Lead Monoxide	PbO	It is used in the manufacture of Lead Stabilizers pottery, Lead glass, paints, enamels, and inks.
Lye/ Caustic soda	Sodium hydroxide	$NaOH$	It is used in manufacture of pulp and paper, alumina, soap and detergents.
Marsh Gas	Methane	CH_4	It is poisonous gas mainly found in mines.
Methyl ethyl ketone/ Butanone	Butan-2-one	C_4H_8O	It is mainly used in paint formation.
Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$	It is used as an antacid to relieve indigestion, sour stomach, and heartburn.
Mohr's Salt	Ammonium iron (II) sulphate	$(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$	It is used to increase the shelf life of iron and reduce oxidation.
Aqua Fortis/ Spirit of Niter	Nitric acid	HNO_3	It is used for production of explosives, and fertilizers.
Muriatic acid	Hydrochloric acid	HCl	It is used as a food additive, and in gelatin production, Leather processing, etc.
Oil of Vitriol	Sulphuric acid	H_2SO_4	It is use for making fertilizer.

Winter green oil	Methyl salicylate	C8H8O3	It is used in traditional medicine for the bacterial infections.
Phosgene	Carbonyl dichloride	COCl2	It is used to make plastics and pesticides.
Plaster of Paris	Calcium sulphate hemihydrate	CaSO4.1/2H2O	It is used as a protective coating on walls and ceilings.
Potash Alum/Potassium Alum	Potassium aluminium sulphate dodecahydrate	KAl(SO4)2·12H2O	It is commonly used in water purification, leather tanning, dyeing.
Rochelle Salt	Potassium sodium tartrate	KNaC4H4O6.4H2O	It is used in manufacturing of pectins and gelatines.
Quartz	Silicon dioxide (Silica)	SiO2	It is used as oscillators in radios, watches.
Quick Silver	Mercury	Hg	It is used in thermometers, thermostats, barometers
Table Salt/Common Salt	Sodium chloride	NaCl	It is commonly used as a condiment and food preservative.
2-propanol/ Propan-2-ol	Isopropyl alcohol	(CH3)2CHOH	It is used to treat tremors, chest pain, hypertension, heart rhythm disorders, and other heart or circulatory conditions.
Sal ammoniac	Ammonium chloride	NH4Cl	It is used as cleaning agent in several industrial processes.
Saltpetre	Potassium nitrate	KNO3	It is used as a common food preservative.
Slaked Lime	Calcium hydroxide	Ca(OH)2	It is used to form mortars, plasters, and cements.
TNT	Trinitrotoluene	C6H2(NO2)3CH3	It is used in used in military shells, bombs, and grenades.

Vermilion/ Cinnabar	Mercuric sulphide	HgS	It is used as a pigment, and carve it into jewelry and ornaments.
Washing Soda	Sodium carbonate decahydrate	Na2CO3. 10H2O	It is used in commercial detergent mixtures to treat hard water.
White Vitriol	Zinc Sulphate	ZnSO4.7H2O	It is used for preserving skins and wood.

Important salt

Common salt :	It is used in large amount as table salt. It is used for preservation of food in pickling. It is used to preserve raw hides. It is used in the manufacture of chemicals such as chlorine and caustic soda (NaOH).
Baking soda :	It is used for making baking powder. It is used for preparing antacid tablets, for controlling acidity of stomach.
Washing soda :	It is used by washermen to wash clothes. It is used in manufacture of glass and caustic soda. It is used in fire extinguishers. It is used in the manufacture of detergents.
Chile saltpetre :	It is used in the manufacture of nitric acid, gun powder and fire works.
Nitre :	It is used as a fertiliser. It is also used in the manufacture of gun powder, fire works and nitric acid.
Smelling salt :	It provides relief from common cold.
Epsom salt :	It is used as a laxative by patients suffering from constipation.
Green vitriol:	It is used in making blue black inks and in curing leather.
Blue vitriol :	It is used as fungicide in agriculture. It is used in dyeing and printing industry. It is used in electroplating copper metal.
Plaster of Paris :	It is used for making statues. It is used for setting broken bones.
Potash alum :	It helps in rapid settling of suspended impurities in water. It is used for the purification of water.

Sodium Iodate	Iodised salt to prevent Goitre's disease.
Sodium Carbonate	Washing soda, manufacturing of glass, etc.
Sodium Benzoate	As a food preservative for pickles
Potassium Nitrate	As a fertilizer giving both K & N to the solid, gun powder, match sticks etc.
Calcium Phosphate	Fertilizer
Alum	Purification of water, dyeing industry, antiseptic, aftershave
Vulcanisation of Rubber	Vulcanisation is a process of treating the natural rubber with sulphur or some compound of sulphur under heat.
Rayon	Synthetic fibre obtained from cellulose

Fibre monomer and uses

FIBRE	Monomers	Uses
Nylon-66,	Adipic acid +hexamethylene diamine	In making bristles for brushes, synthetic fibres, parachutes.
Terylene	Ethylene glycol and terephthalic acid	For making wash and wear fabrics, tyre cords, safety belts, tents etc.
Kevlar	Terephthalic acid + 1,4-diamino benzene	For making bulletproof vests.

Lexan or polycarbonate	Diethyl carbonate + *bis*-phenol-A	In making bulletproof windows.
Polyurethanes	Toluene diisocyanate + ethylene glycol	For making washable and long lasting mattresses, cushions.